DO NOW — RECALL

Lesson 148 · Network Hardware: Switches, Routers and

Name: _____ Class: _____ Date: _____

1. What does a firewall generally use to permit or block a connection?

2. What data structure follows FIFO?

3. What is the result of an AND gate with inputs 1 and 0?

4. What is a foreign key used for?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A policy of rules, for example addresses, ports or connection state.

2. Model answer

A queue.

3. Model answer

0.

4. Model answer

Referencing a candidate key in another relation.

By the end, you can…

• Identify the role of switches, routers, wireless access points, NICs and repeaters.

• Follow a frame within a LAN and a packet between LANs, distinguishing MAC and IP use.

• Select network hardware for a realistic school network and justify its placement.

Vocabulary — know these terms

switch — Connects local devices and forwards data-link frames.

router — Forwards network-layer IP packets between networks.

access point — Connects wireless devices to a LAN.

NIC — Network interface controller for network connectivity.

NEW LEARNING

Read, annotate and explain

Local connections

A switch forwards Ethernet frames using learned MAC addresses.

Across networks

Router consults IP destinations and routing information.

Wireless access

Access point bridges wireless stations onto a network.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Two school laptops on one wired LAN share a file; one then accesses a website on another network. Trace the hardware.

Laptop and switch

Identify the facts and make an initial prediction.

Router/default gateway

Explain the mechanism or trace the intermediate stage.

Remote network

Within the LAN, NICs form frames and the switch forwards using MAC addresses. To reach another IP network, traffic is sent to a router/default gateway that forwards packets. The gateway rewrites the link framing per next hop.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Differentiate switch, router and access point with their forwarding roles. [6]

Question 2

For a PC accessing a website, explain why both MAC and IP addresses are useful. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Design a network for two labs and a library with wireless devices and an internet connection. Label device placement and traffic path. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A basic Ethernet switch chooses internet routes using destination website names.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.